

Sydney Girls High School



MATHEMATICS

YEAR 11

2021 (online) Assessment Task 3

Reading time: 5 minutes

Writing time: 60 minutes

Total marks: 40

Topics: Functions, Calculus, Trigonometric functions, Exponential and Logarithmic Functions and Statistical Analysis (excluding Discrete probability distributions).

General Instructions:

- There are four (4) questions which are of equal value.
- Attempt all questions.
- Show all necessary working. Marks may be deducted for badly arranged work or incomplete working.
- Start each question on a new page.
- Write on one side of the paper only.
- Diagrams are NOT to scale.
- Board-approved calculators may be used.
- Write using a black or blue pen.
- A correct answer without working will be awarded a maximum of 1 mark.

Question 1:**(10 Marks)**

- a) Solve:
- i) $|7 - 6x| = 5$ 2
- ii) $\left(\frac{1}{3}\right)^{x+1} = 27$ 2
- b) Find the centre and radius of $x^2 - 2x + y^2 + 4y = 1$. 2
- c) If $\frac{\sqrt{6}+1}{\sqrt{3}+\sqrt{2}} = a\sqrt{3} + b\sqrt{2}$, find the values of a and b . 2
- d) Find the equation of the tangent to the curve $y = x^2 - 3x$ at $x = 1$. 2

Question 2: Start writing on a new page**(10 Marks)**

- a) Find $P(X \mid Y)$ if $P(X \cap Y) = 0.2$, $P(X) = 0.3$ and $P(Y) = 0.4$. 1
- b) Differentiate the following:
- i) $y = (5x - 7)^5$ 1
- ii) $y = \frac{3x - 2\sqrt{x}}{\sqrt{x}}$ 2
- iii) $y = \frac{e^{3x} - 5}{x + 1}$ 2
- c) Given the points A (10,2) and B (2,8) find the equation of the perpendicular bisector of AB. 2
- d) Solve $2 \cos \theta = -1$ for $0 \leq \theta \leq 2\pi$, giving exact solutions. 2

Question 3: Start writing on a new page**(10 Marks)**

a) Sketch the graph of $y = \operatorname{cosec} x$ for $0^\circ \leq x \leq 360^\circ$. 1

b) If $\sin A = -\frac{1}{3}$ and $\tan A < 0$. Find $\sec A$. 2

c) Sketch the following, showing all the key features:

i) $y = -\sqrt{25 - x^2}$ 1

ii) $y = 2 - \frac{1}{x+1}$ 2

d) Angela guesses three questions in her multiple choice test, which has four options per question. Find the probability that Angela gets:

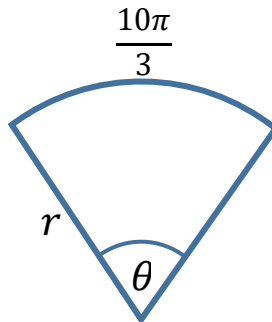
i) At least one correct answer 1

ii) Only one correct answer 1

e) Prove $\frac{\cos \theta}{1 + \sin \theta} + \tan \theta = \sec \theta$. 2

- a) The diagram shows a sector with radius r and angle θ where $0 < \theta \leq 2\pi$.

The arc length is $\frac{10\pi}{3}$.

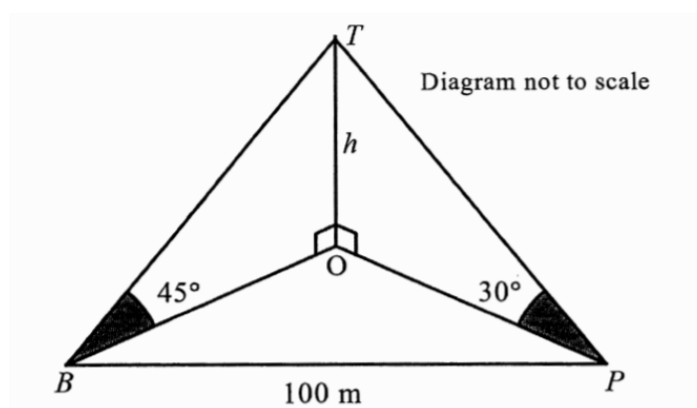


- | | | |
|-----|--|---|
| i) | Find the value of θ and hence, show that $r \geq \frac{5}{3}$. | 2 |
| ii) | Calculate the area of the sector when $r = 4$. | 2 |

- b) Consider the function $f(x) = e^x$ and $g(x) = \ln(x - 2)$.

- | | | |
|-----|---|---|
| i) | Find $f(g(x))$. | 1 |
| ii) | Find the range of $f(g(x))$ in interval notation. | 1 |

- c) A surveyor stands at a point P, which is due east of the tower OT, of height h metres.
 The angle of elevation of the top of the tower T from P is 30° . The surveyor then walks 100 metres to point B, which is on a bearing of 150° from the foot of tower O.
 The angle of elevation of the top of the tower from B is now 45° .



- | | | |
|-----|---|---|
| i) | Express the length of OP in terms of h . | 1 |
| ii) | Find the height of the tower.
Answer correct to 1 decimal place. | 3 |

Question 1

(a)(i) $|7 - 6x| = 5$

$$7 - 6x = 5 \quad \text{or} \quad -(7 - 6x) = 5$$

$$6x = 2 \quad -7 + 6x = 5$$

$$x = \frac{1}{3}$$

$$6x = 12$$

$$x = 2$$

(ii) $\left(\frac{1}{3}\right)^{x+1} = 27$

$$3^{-x-1} = 3^3$$

$$-x - 1 = 3$$

$$x = -4$$

(b) $x^2 - 2x + y^2 + 4y = 1$

$$(x - 1)^2 - 1 + (y + 2)^2 - 4 = 1$$

$$(x - 1)^2 + (y + 2)^2 = 6$$

Centre $(1, -2)$

Radius $\sqrt{6}$

(c) $\frac{\sqrt{6} + 1}{\sqrt{3} + \sqrt{2}} \times \frac{\sqrt{3} - \sqrt{2}}{\sqrt{3} - \sqrt{2}} = \frac{3\sqrt{2} - 2\sqrt{3} + \sqrt{3} - \sqrt{2}}{3 - 2}$

$$= 2\sqrt{2} - \sqrt{3}$$

$$a = -1 \text{ and } b = 2$$

For getting at least one of the completed squares correct.

(d) $y = x^2 - 3x$

$$\frac{dy}{dx} = 2x - 3$$



$$m_T = -1 \text{ at } x = 1$$

$$(1, -2)$$

$$y + 2 = -1(x - 1)$$

$$y = -x - 1$$



Question 2

$$\begin{aligned} \text{(a)} \quad P(X|Y) &= \frac{P(X \cap Y)}{P(Y)} \\ &= \frac{0.2}{0.4} \\ &= \frac{1}{2} \end{aligned}$$

$$\text{(b)(i)} \quad \frac{dy}{dx} = 5(5x - 7)^4 \times 5$$

$$\frac{dy}{dx} = 25(5x - 7)^4$$

$$\text{(ii)} \quad y = 3\sqrt{x} - 2$$

$$\frac{dy}{dx} = \frac{3}{2}x^{-\frac{1}{2}}$$

$$\frac{dy}{dx} = \frac{3}{2\sqrt{x}}$$

$$\text{(iii)} \quad \frac{dy}{dx} = \frac{3e^{3x}(x+1) - 1(e^{3x} - 5)}{(x+1)^2}$$

$$\frac{dy}{dx} = \frac{3xe^{3x} + 3e^{3x} - e^{3x} + 5}{(x+1)^2}$$

$$\frac{dy}{dx} = \frac{3xe^{3x} + 2e^{3x} + 5}{(x+1)^2}$$

$$\text{(c)} \quad m_{AB} = \frac{8-2}{2-10}$$

$$m_{AB} = -\frac{3}{4}$$

$$m_{\perp} = \frac{4}{3}$$

$$MP_{AB} = (6, 5)$$

$$y - 5 = \frac{4}{3}(x - 6)$$

$$3y - 15 = 4x - 24$$

$$4x - 3y - 9 = 0$$

$$\text{(d)} \quad \cos \theta = -\frac{1}{2}$$

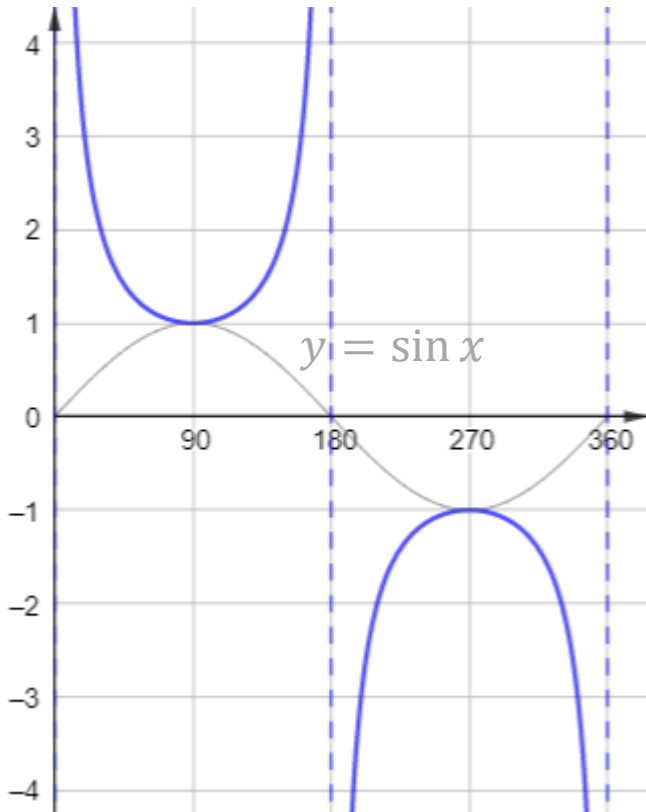
$$\theta = \pi \pm \frac{\pi}{3}$$

$$\theta = \frac{2\pi}{3}, \frac{4\pi}{3}$$

Question 3

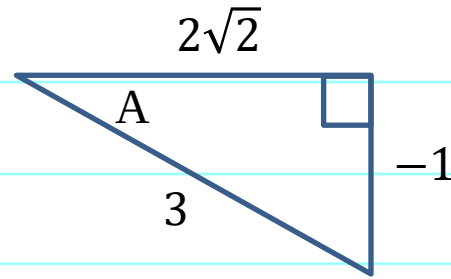
$$y = \operatorname{cosec} x$$

(a)



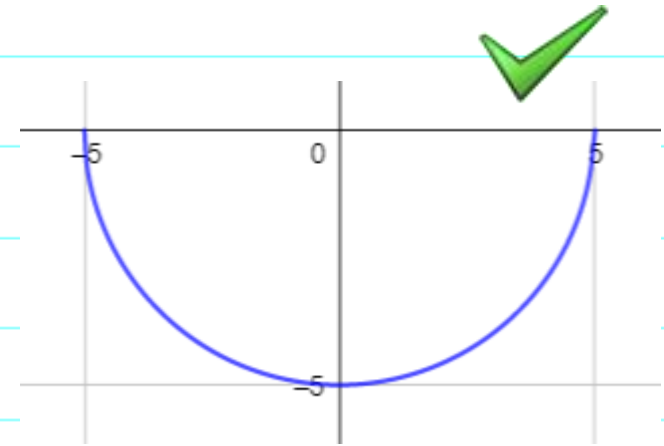
(b) $\sin A = -\frac{1}{3}$ and $\tan A < 0$

So A is in the 4th quadrant.

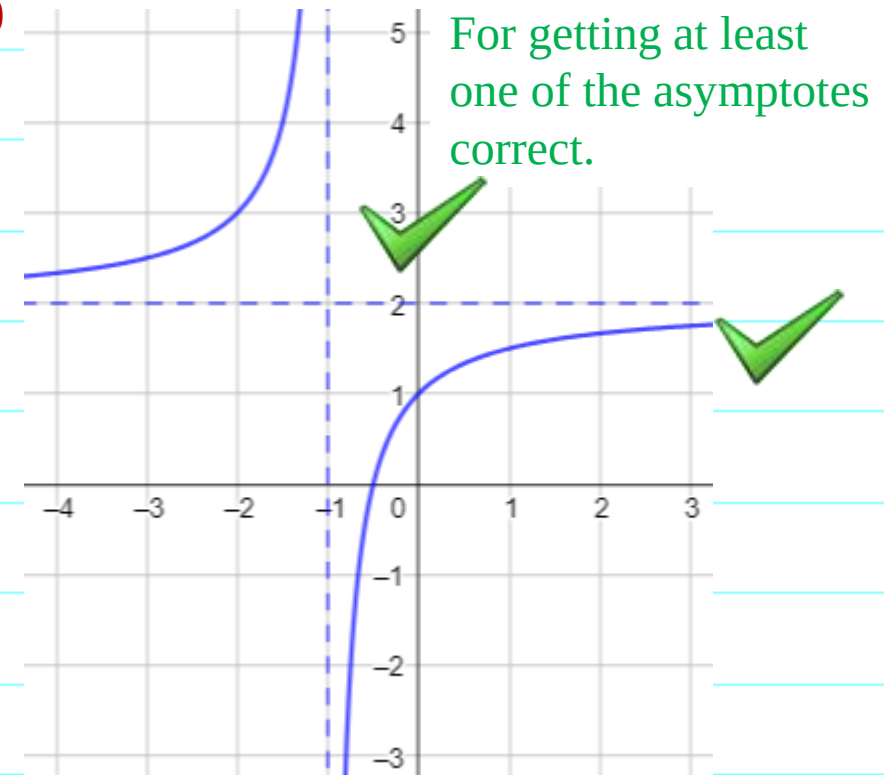


$$\begin{aligned} \sec A &= \frac{1}{\cos A} \\ &= \frac{3}{2\sqrt{2}} \\ &= \frac{3\sqrt{2}}{4} \end{aligned}$$

(c)(i)



(ii)



(d)(i) $P(\text{at least one correct}) = 1 - P(\text{three incorrect})$

$$= 1 - \left(\frac{3}{4}\right)^3$$

$$= \frac{37}{64}$$

(ii) $P(\text{Only 1st correct}) + P(\text{Only 2nd correct}) + P(\text{Only 3rd correct})$

$$= \left(\frac{1}{4} \times \frac{3}{4} \times \frac{3}{4}\right) + \left(\frac{3}{4} \times \frac{1}{4} \times \frac{3}{4}\right) + \left(\frac{3}{4} \times \frac{1}{4} \times \frac{3}{4}\right) = 3 \times \frac{3^2}{4^3}$$

$$= \frac{27}{64}$$

(e) $LHS = \frac{\cos \theta}{1 + \sin \theta} + \frac{\sin \theta}{\cos \theta}$

$$= \frac{\cos^2 \theta + \sin \theta + \sin^2 \theta}{(1 + \sin \theta) \cos \theta}$$

$$= \frac{1 + \sin \theta}{(1 + \sin \theta) \cos \theta}$$

$$= \frac{1}{\cos \theta}$$

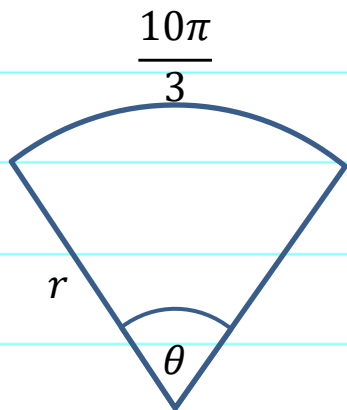
$$= RHS$$

Question 4

(a)(i) $l = r\theta$

$$\frac{10\pi}{3} = r\theta$$

$$\theta = \frac{10\pi}{3r}$$



(b)(i) $f(g(x)) = e^{\ln(x-2)}$ where $x - 2 > 0$

$$= x - 2 \text{ for } x > 2$$

To get the mark the domain must be included.

(ii) $f(g(x)) > 0$

$$(0, \infty)$$

Given that $\theta \leq 2\pi$

Thus $\frac{10\pi}{3r} \leq 2\pi$

$$\frac{10\pi}{6\pi} \leq r$$

$$r \geq \frac{5}{3}$$

(ii) Area = $\frac{1}{2}r^2\theta$

$$= \frac{1}{2} \times 4^2 \times \frac{10\pi}{3 \times 4}$$

$$= \frac{20\pi}{3}$$

(c)(i) $\tan 30^\circ = \frac{h}{OP}$

$$OP = h\sqrt{3}$$

(ii) $\tan 45^\circ = \frac{h}{OB}$

$$OB = h$$

$$100^2 = h^2 + 3h^2 - 2\sqrt{3}h^2 \cos 60^\circ$$

$$100^2 = h^2(4 - \sqrt{3})$$

$$h = \frac{100}{\sqrt{4 - \sqrt{3}}} \approx 66.4 \text{ m}$$

